

Research Profile

- Focus Design-led researcher on embodied AI tutors and multimodal instruction, linking language-gesture science to AI-enhanced teaching.
- Training PhD frameworks for meaningful instructional gestures in embodied AI teachers (*IJHCI*, CSCW).
- Prototypes Two skeletal-rigged embodied AI-teacher lecture prototypes; also health and social apps.
- Teaching 6 years university lecturing; PolyU postdoc since 2024; supervised student prototypes of interactive platforms and virtual systems.
- Keywords **AI in Education; Embodied AI Tutors; Co-speech Gesture; Embodied Pedagogical Agents; Language-Gesture Integration; Multimodal Interaction; ITS.**

Current Research

- Corpus With PolyU LST and South China Normal Univ. (Prof. Suiping Wang), collecting naturalistic dyadic dialogue in word-guessing tasks to build a multimodal corpus of coordinated speech, gesture, gaze, and fNIRS.
- Pipeline Integrating gesture, speech, gaze, fNIRS, and related cues with CS partners to investigate how humans coordinate multimodal information, **informing more trustworthy agents.**
- Platform Scalable tools for multimodal analytics and embodied AI tutor deployment (Education 4.0 / intelligent tutoring).

Research and Teaching Experience

- 2026–present **Postdoctoral Fellow, Chen Si Lab, Language Science and Technology (LST), PolyU**
Multimodal dialogue, gesture, and fNIRS collection; language-gesture integration; grant proposal development; AI-enhanced instruction experiments.
- 2025–2026 **Postdoctoral Fellow, Brain, Language and Computation Lab (BLC Lab), LST, PolyU**
Multimodal data analysis; dialogue gesture corpus development; grant proposal development; AI-enhanced instruction experiments.
- 2019–2022 **Teaching Assistant, PolyU School of Design (MDes and BA)**
Interaction design, UX, concept workshop, and research and analysis studios.
- 2017–2023 **Lecturer, Visual Communication Design (BA), Chongqing Univ. of Education**
Multimedia design, animation, UX/UI, and design studio teaching (6 years).

Publications

- 2026 **Grounding Pedagogical Intent in Embodied AI Teachers: A Human-Centered Framework for Designing and Evaluating Instructional Gestures.**
Lai Wei, Pangrui Xing, Kenny K. N. Chow, Stephen Jia Wang, International Journal of Human-Computer Interaction (SSCI, Q1; IF 4.9)
- 2026 **CapSenseBand: Sustaining cross-disciplinary creativity when stitches must meet signals.**
Sark Pangrui Xing, Hongci Hu, Lai Wei, Le Fang, Ziqian Bai, Kinor Shou-xiang Jiang, Stephen Jia Wang, Proc. Creativity&Cognition (CC '26) (Rank A HCI)
- 2024 **Motion tracking of daily living and physical activities in healthcare: A systematic review from designers' perspective.**
Lai Wei, Stephen Jia Wang, JMIR mHealth and uHealth (SCIE, Q1; IF 6.2)
- 2024 **Developing cue-behavior association for habit formation: A qualitative study to explore the role of avatar in hypertension.**
Yujie Zhu, Yonghao Long, Lai Wei, Yaqi Zhang, Zhengtao Ma, Kun-Pyo Lee, Lin Zhang, Stephen Jia Wang, Digital Health (SCIE, SSCI, Q1; IF 3.9)
- 2023 **When Gestures and Words Synchronize: Exploring A Human Lecturer's Multimodal Interaction for the Design of Embodied Pedagogical Agents.**
Lai Wei, Kenny K. N. Chow, Companion Proc. CSCW '23 (Rank A HCI; CCF-A)
- 2023 **How students perceive lecturers' gestures? An exploration in gesture-meaning matching toward embodied pedagogical agent design.**
Lai Wei, Kenny K. N. Chow, IASDR '23 (Rank A Design)
- 2022 **Interface, pedagogical agents and chatbox in virtual learning environments: An eye-tracking experiment.**
Lai Wei, Kenny K. N. Chow, IASDR '22 (Rank A Design)

- 2026 **Gender differences in students' perceptions and outcome of feedback from embodied AI instructors.**
Mingjun Zhai, Lai Wei, Zhe Gao, Ping Li, CogSci 2026 (poster; Rank A Cognitive Science)
- 2025 **The roles of gestures in dyadic interaction: Associations with effective communication, emotion regulation and individual traits.**
Lai Wei, Mingjun Zhai, Suiping Wang, Ping Li, ISGS '25 (poster; Rank A Gesture Studies)
- 2025 **From human lecturers to pedagogical agents: Investigating word–gesture alignment for instructional gesture design.**
Pangrui Xing, Lai Wei, Kenny K. N. Chow, ISGS '25 (poster; Rank A Gesture Studies)
- 2025 **How shared interactive experiences reshape the semantic representations of abstract concepts.**
Mingjun Zhai, Lai Wei, Ping Li, CogSci 2025 (poster; Rank A Cognitive Science)

Under review

- Who Does the Embodied AI Teacher Gesture For? A Lecturer–Student Interview Study of Teaching Gestures.**
Lai Wei, Sark Pangrui Xing, Kenny K. N. Chow, CHI 2027
- A Corpus Study of Gesture Rates among Undergraduate Instructors for Embodied AI Teachers.**
Lai Wei, Sark Pangrui Xing, Kenny K. N. Chow, CHI 2027
- Delivering Personalized LLM-Generated Feedback Through Embodied Pedagogical Agents: The Roles of Visual Realism and Perceived Humanness.**
*Mingjun Zhai, Lai Wei, Zhe Gao, Jiaxin Chen, Yongqing Li, Ping Li, Computers & Education Beats and pointing with English *not*, *n't* and *no* across grammatical class and medium.*
- Lai Wei, Pangrui Xing, Kenny K. N. Chow, Gesture Studies*
- Culturally grounded but socially malleable: Cooperative dialogue recalibrates abstract concept representations.**
Mingjun Zhai, Lai Wei, Yingying Peng, Ping Li, Communications Psychology (Nat. Portfolio)

In preparation

- When First Impressions Meet Feedback: Appearance, Valence, and Evaluations of AI Teachers in AI-Enhanced Writing Instruction.** (target: *Computers & Education*)

Education

- 2019–2024 **PhD, School of Design, The Hong Kong Polytechnic University**
 Dissertation: *Animated Gestures: Pedagogical Agents' Meaningful Instructional Gesture Design for Effective Educational Information Presentation.*
 Supervisor: **Kenny K. N. Chow**; examiners: **Wang Minjuan, Dragan Gasevic**
- 2013–2015 **MA, Central Saint Martins, University of the Arts London, London, UK**
 Character animation, illustration, and film producing.
- 2008–2012 **BA, Animation and Digital Arts, Communication University of China**
 Human figure animation and movement design; digital arts.

Grants and Awards

- 2025 **Postdoctoral Matching Fund, PolyU** **HK\$794,000**
- 2025 **Co-Principal Investigator, MOE Humanities and Social Sciences Project** **RMB 100,000**
 From Myth to Gameplay: The transmediation of symbols and construction of identity in gamified worlds.
- 2023 **Additional Top Conference Grant, PolyU** **HK\$20,000**
- 2020 **Co-Principal Investigator, Municipal Education Commission of Chongqing** **RMB 50,000**
 Data visualization-based learning analytics system.
- 2019 **Teaching Postgraduate Studentship, PolyU** **HK\$593,000**
- 2019 **PI, CQUE Teaching Research Grant** **RMB 3,000**
- 2023 **Invited presentation, ChinaVis 2023 (Visualization and Visual Analytics)**
- 2021 **Best Presenter Award, 12th PhD Colloquium on Design Research, Tsinghua University**
- 2012–2016 **Animation festival selections and awards**
 2nd Place, Animex Awards (2016) and World Animation Celebration (2015); official selections at British Animation Awards, Taiwan International Children's Film Festival, and AnimaSyros.

University Service

- Peer review Referee for *Brain and Language*, *IEEE Transactions on Learning Technologies*, *Education and Information Technologies*, *JMIR mHealth and uHealth*, *European Journal of Education*, *ACM CSCW*, and *IASDR* (2023–2026).
- 2021–present **Senior Hall Tutor, Hung Hom Halls (LS Hall), PolyU**
 Coordinating hall tutors, student mentoring, and hall educational activities.

Skills and Collaborators

- Methods Mixed methods; lab experiments; eye-tracking; discourse and multimodal analysis.

Tools Pedagogical agent prototyping (LLM-assisted, skeletal rigging, and other pipelines); EyeLink, ELAN, CLAN, Praat, NVivo, R, SPSS; Adobe, Cinema 4D, Figma.

Collaborators Ping Li (HKUST); Suiping Wang (SCNU); Kenny K. N. Chow (HKBU); Wang Minjuan (EdUHK); Stephen Jia Wang (PolyU Design); Chen Si (PolyU LST).